

Modeling the bathtub shape hazard rate function in terms of reliability

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Abstract

In this paper, a general form of bathtub shape hazard rate function is proposed in terms of reliability. The degradation of system reliability comes from different failure mechanisms, in particular those related to (1) random failures, (2) cumulative damage, (3) man–machine interference, and (4) adaptation. The first item is referred to the modeling of unpredictable failures in a Poisson process, i.e. it is shown by a constant. Cumulative damage emphasizes the failures owing to strength deterioration and therefore the possibility of system sustaining the normal operation load decreases with time. It depends on the failure probability, $1 - R$. This representation denotes the memory characteristics of the second failure cause. Man–machine interference may lead to a positive effect in the failure rate due to learning and correction, or negative from the consequence of human inappropriate habit in system operations, etc. It is suggested that this item is correlated to the reliability, R , as well as the failure probability. Adaptation concerns with continuous adjusting between the mating subsystems. When a new system is set on duty, some hidden defects are explored and disappeared eventually. Therefore, the reliability decays combined with decreasing failure rate, which is expressed as a power of reliability. Each of these phenomena brings about the failures independently and is described by an additive term in the hazard rate function $h(R)$, thus the overall failure behavior governed by a number of parameters is found by fitting the evidence data. The proposed model is meaningful in capturing the physical phenomena occurring during the system lifetime and provides for simpler and more effective parameter fitting than the usually adopted ‘bathtub’ procedures. Five examples of different type of failure mechanisms are taken in the validation of the proposed model. Satisfactory results are found from the comparisons. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Dynamic reliability; Bathtub curve; Hazard rate

1. Introduction

Understanding the dynamic behavior of system reliability, R , becomes an important issue in either scheduling the maintenance activities or dealing with the improvement in the revised system design. In doing so, the hazard rate function should be addressed. Bathtub curve is usually adopted to represent the general trend of hazard rate function, h . Many studies [1–12] were concentrated on the study of time-dependent relation of h or/and R for some typical relations. However, these studies focus their effort on depicting the geometric shape of bathtub curve with less discussion on the physical meanings. In general, a system adjusts its performance continuously according to the capability concerned with (1) adaptation between mated subsystems and system with surroundings, and (2) resistance to the cumulative damage. These two mechanisms effect system performance simultaneously, the former dominates behavior in the infant mortality stage, the latter

does in the wear out stage, see Fig. 1. For example, the mating gears shape their geometries during the initial transmission period (adapting to each other) meanwhile the potential of failure in the transmission appears gradually due to either the tooth broken or serious wear in the mating surfaces (both resulting from cumulative damage). In conventional approach, two-parameters Weibull distribution is taken to fit data with either the running-in period or the wear out stage. For more complicated situation in which the data cover the running-in, useful life as well as the wear out stage, mixed-Weibull distribution with three-set parameters and two transition points (totally eight-parameters) should be considered. Therefore, the fitting becomes a tremendous work.

The purpose of this paper is to develop a model concerned with the behavior of hazard rate in the whole range of bathtub curve, based on the failure mechanisms. The relation between the hazard rate and the reliability of a system follows the definition

$$h(t) = -\frac{1}{R(t)} \frac{dR}{dt}. \quad (1a)$$

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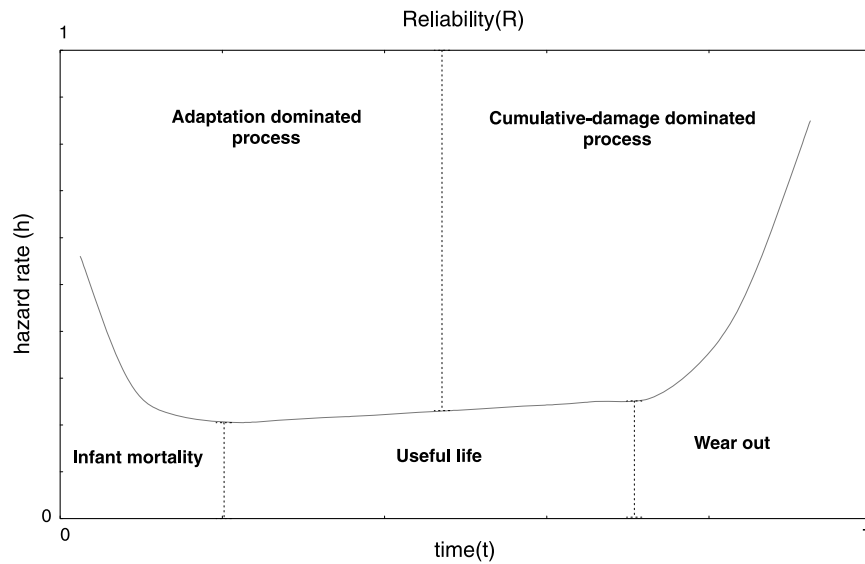


Fig. 1. The general behavior of hazard rate vs. time or reliability.

Usually the reliability decreases monotonically with time, thus there is a one-to-one correspondence between R and t , i.e. the hazard rate function can also be expressed as:

$$h(t) = -\frac{1}{R} \frac{1}{dt/dR} = f(R). \quad (1b)$$

Thus, instead of the usual procedure of estimating $h(t)$, we propose to fit the relationship of $h(R)$ based on the available data. The change of the expression $h(t)$ to $h(R)$ has certain advantages. First, the equation of dynamic reliability takes an autonomous form; particularly it belongs to a general type of logistic equation encountered very often in ecological science [13]. Therefore good experience can be guided from these studies. Secondly, the hazard rate is investigated in finite domain $[1,0]$ as comparing with that in infinite domain of time sequence. Sometimes the fitting of $h(t)$ has to concern whether or not the collected data in time domain are enough to depict the actual circumstances, but this is not beared in taking the fitting of $h(R)$. In the past years, the following relations were studied

$$h = A_0 + A_1(R_0 - R) + \text{higher order terms}, \quad (2a)$$

or

$$\frac{dh}{dt} = a_1(R_0 - R) + \text{higher order terms}, \quad (2b)$$

for dynamic reliability of material fracture due to crack growth [14] and development of new mechanical product with modified function requirements [15]. Neglecting the higher order terms and dividing Eq. (2b) by Eq. (1a) lead to an integrable equation for $h(R)$. The result is

$$h(R) = \left[h_0^2 + 2a_1R_0 \ln\left(\frac{R_0}{R}\right) - 2a_1(R_0 - R) \right]^{1/2}, \quad (2c)$$

where R_0 and h_0 are the initial conditions of reliability and hazard rate, respectively. Eq. (2a) is called the algebraic

equation model (AE model), Eq. (2b) or (2c) the differential equation model (DE model). The results show that both relations agree well with simulations. The difference between these two models is that the hazard rate approaches to a constant in AE model rather than infinite considered with DE model. However, no evidence is found about the behavior of hazard rate, as the system reliability tends to zero. The proposed reliability-dependent models have the merit to demonstrate the characteristics of cumulative damage [16], since it shows that the hazard rate increases as the system strength deteriorates. This is found indirectly in the two-parameters Weibull representation of system reliability. Meanwhile the dynamic reliability based on Eq. (1a) without higher order terms can be integrated in a closed form, i.e. this simplifies the fitting. Moreover the AE model was considered for other mechanical failures, such as fatigue based on $S-N$ curve [17–19], sliding wear of carburized steel [20], material creep failure [21], cutting tool's wear [22], etc.

In this paper the relation $h(R)$ is further extended to include other effects: such as (1) adaptive mechanism, which plays an important role on the system behavior at the beginning stage, particularly when system mating with its environment, and (2) man-machine interference (this is a general term describing the effects due to human learning, training consequence, some habit in the system operation, or the like) which may effect the system performance, especially within the useful life period. Finally, five examples of different type of failure mechanisms are taken in the model validation.

2. Reliability-dependent bathtub hazard rate function

During the system service, the variation of hazard rate comes from different sources including (1) unpredictable

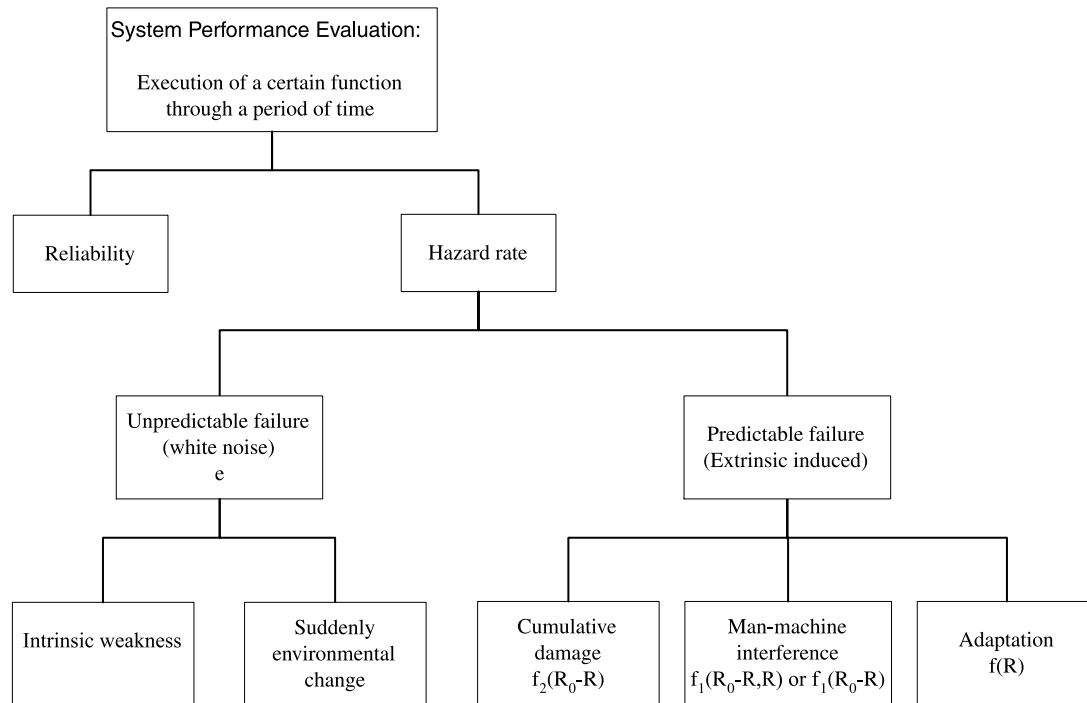


Fig. 2. Classification of failure mechanisms.

failures due to the intrinsic weakness embedded within the system or/and unexpected environmental changes and (2) extrinsic-induced degradation, see Fig. 2 for details. The physical explanations of the proposed reliability-dependent relations are described below

(a) *White noise*: it concerns with unpredictable failure occurrence either from the sudden increase of applied load due to environmental variations or the unpredictable intrinsic weakness embedded within a new system [15]. Such conditions will make the system fail randomly during the service life. Suppose the occurrence of failures follows a Poisson process, i.e. this type of failure represented by a constant, e .

(b) *Cumulative damage*: it denotes that the strength weakening occurs continuously due to the applied stress, typical example is shown in fatigue phenomenon [14,19]. This type hazard rate emphasizes the memory characteristics of strength degradation. Since $R_0 - R$ denotes the failure probability, thus the relation with reliability is further modified as [23]

$$h_c = c(R_0 - R)^n, \quad c > 0, \quad n > 0. \quad (3a)$$

The parameter c relates to substantial decrease in reliability as the cumulative damage becomes significant in the failure possibility, hence, it could be thought as the coefficient of strength degradation [16]. The exponent n denotes the beginning of obvious change in reliability, thus, it refers to the memory characteristics of damage [16]. Without loss of generality, put $R_0 = 1$, the above relation becomes

$$h_c = c(1 - R)^n, \quad c > 0, \quad n > 0, \quad (3b)$$

(c) *Man-machine interference*: it may result in a positive or negative effect in the system failure. Usually the hazard rate would change slightly in the useful life period. The explanation of such tiny variation is very complicated. Logical learning corrects mistakes. Assume the chance to have a mistake depends on the failure probability, i.e. $1 - R$, and meanwhile the ability of recovery through learning can be measured by a power of reliability. Since human learning and machine failure behavior interacts with each other, the hazard rate of this type can be proposed in the form

$$h_m = lR^{m'}(1 - R)^{n'}, \quad m', n' > 0, \quad (4a)$$

where the learning coefficient, l , can be either positive or negative, m' and n' represent the learning and recovery exponents, respectively. For positive l , Eq. (4a) represents that the human operation (it may come from unqualified procedures) leads the system in a worse situation with negative learning, and thus the hazard rate rises. This is observed frequently that the hazard rate of man-machine system increases slightly owing to not well-trained operations. While for negative l , it denotes that the learning effects adjust the operation in right track and thus the hazard rate reduced. In some cases, the recovery characteristics may not be sensitive to the reliability, h_m can be further simplified to

$$h_m = l(1 - R)^{n'}. \quad (4b)$$

Eq. (3b) is distinguishable from Eq. (4b), the former depicts the behavior of failure rate in the wear out stage while the latter indicates the slight increase of failure rate

in useful life stage, i.e. $n > n'$ and $c > l > 0$, see example 4 in the Section 3. However, few examples can be found in the literature to show the improvement of failure rate of a complicated system due to human's correction from learning.

(d) *Adaptation*: it concerns with the process of improvement in the failure rate as the adaptation between mating components or subsystems. However such mechanism reduces the components strength when the mating becomes smooth, thus the reliability decays with decreasing failure rate, i.e.

$$h_a = aR^m, \quad a > 0, \quad m > 0, \quad (5)$$

where a and m are the adaptation strength and characteristics, respectively.

The difference between type (d) failure with (c) is that, the mistakes in type (d) cannot be learned and corrected. Although the adaptation and white noise both involve with the intrinsic weakness of a new system, they still can be recognized from each other. Adaptation decreases the failure rate during mating. For instance, the roughness of gear surface is improved according to the repeated sliding motion, thus the failure rate becomes smaller in the engagement. On the other hand, for other mechanisms the failure rate may not be reduced but enhanced thereafter with cumulative damage, e.g. material dislocations from manufacturing processes cannot be eliminated but enlarged, which is the kernel (denoted as a part of white noise) of fatigue failure. Before examining the proposed relation, it is assumed that the effects of mentioned mechanisms are independent to each other. By adding e and Eqs. (3b), (4a) or (4b) and (5), the general form of hazard rate function becomes

$$h(R) = e + aR^m + c(1 - R)^n + lR^{m'}(1 - R)^{n'}, \quad (6a)$$

or

$$h(R) = e + aR^m + c(1 - R)^n + l(1 - R)^{n'}. \quad (6b)$$

Suppose that the parameters in the before equation are all positive except l , it might be negative in some occasions. For $l < 0$, a sufficient condition about the magnitude of l to guarantee $h(R) > 0$ can be conducted as follows.

Since a and c are positive it should have

$$h(R) > h^*(R) = e + lR^{m'}(1 - R)^{n'}, \quad \text{for } m' \neq 0, \quad n' \neq 0.$$

$h^*(R)$ has minimum, for $l < 0$, at $R = R^* = m'/(m' + n')$, it is

$$h_m^* = h^*(R^*) = e + l \left(\frac{m'}{m' + n'} \right)^{m'} \left(\frac{n'}{m' + n'} \right)^{n'}.$$

If $h_m^* > 0$, then this implies $h(R) > 0$, thus

$$|l| < e \frac{(m' + n')^{(m' + n')}}{m'^{m'} n'^{n'}}. \quad (7)$$

Substituting $h(R)$ into Eq. (1a) leads to the equation of dynamic reliability as

$$\frac{dR}{dt} = -[e + aR^m + c(1 - R)^n + lR^{m'}(1 - R)^{n'}]R, \quad (8a)$$

or

$$\frac{dR}{dt} = -[e + aR^m + c(1 - R)^n + l(1 - R)^{n'}]R. \quad (8b)$$

To solve the above equations the parameters in $h(R)$ should be identified first from experimental data about $[h, R]$ by existing softwares (e.g. non-linear estimation in Statistica), then the integration of Eq. (8a) or (8b) can be carried out by available numerical packages (e.g. ode45.m in Matlab).

3. Case studies

In this section, five examples are taken to examine the proposed hazard rate model. In the first, the reliabilities are estimated by the probabilities of survivals or $1 - F$ (F is median rank of failure probability) from the given data. The hazard rates are evaluated according to the mean value theorem in calculus, i.e.

$$h_m = h(t_m) = h\left(\frac{t_i + t_{i+1}}{2}\right) = \frac{R(t_i) - R(t_{i+1})}{R\left(\frac{t_i + t_{i+1}}{2}\right)(t_{i+1} - t_i)} \quad (9a)$$

where

$$R\left(\frac{t_i + t_{i+1}}{2}\right)$$

is approximated by

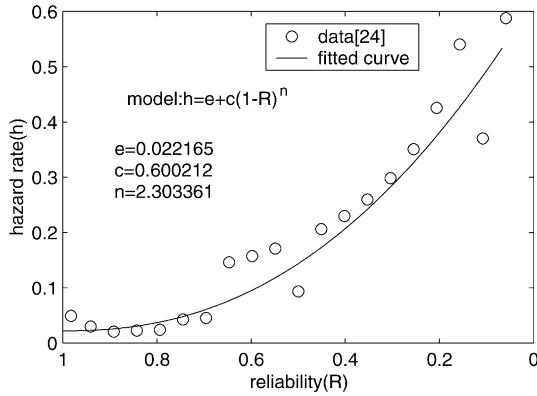
$$R\left(\frac{t_i + t_{i+1}}{2}\right) = \frac{1}{2}[R(t_i) + R(t_{i+1})]. \quad (9b)$$

The results are arranged in time (or distance) sequence with reliability and the corresponding hazard rates. Eq. (6a) or its reduced form Eq. (6b) is fitted to decide the parameters, and then the comparisons between the fitted models and given data are presented in figures.

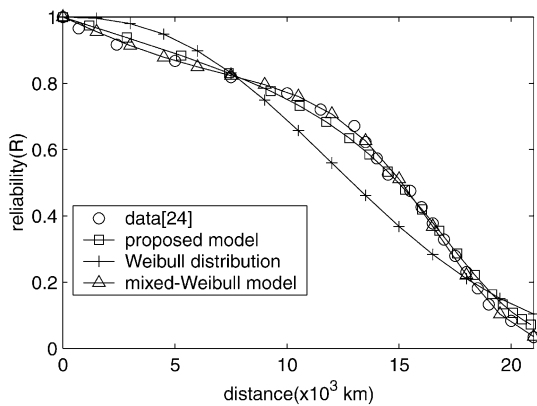
3.1. Case 1

There are reliability data of certain automobile components recorded from some transportation cooperation [24]. Checking the plot of h vs. R in Fig. 3(a), no significant contributions due to adaptation as well as man-machine interference in the hazard rate model are noticed. Thus, Eq. (6a) reduces to

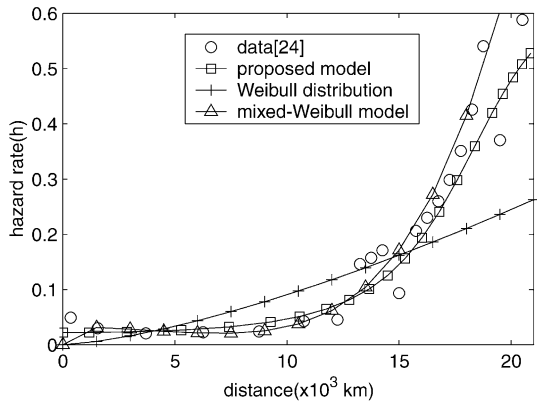
$$h(R) = e + c(1 - R)^n. \quad (10)$$



(a) The fitting of hazard rate vs. reliability



(b) The fitted reliability vs. distance



(c) The fitted hazard rate vs. distance

Fig. 3. The comparison of fitted results with data [24] in case 1.

The fitting shows

$$h(R) = 0.022165 + 0.600212(1 - R)^{2.303361}, \quad (11)$$

with correlation coefficient 0.9636 between Eq. (11) and the data set $[h, R]$ (in the following cases, we also present correlation coefficient between $h(R)$ and data in italic type to denote the fitting accuracies in the examples). Substituting Eq. (11) into Eq. (1a) and integrating R lead to the reliability variation as shown in Fig. 3(b), see the

curve with the symbol within quotes. The correlation coefficient between the reliability curve and the data of R is 0.9973. To compare this approach with two-parameters Weibull distribution model, the same data are used and it results in

$$R(t) = \exp\left(-\left(\frac{t}{15.00571}\right)^{2.4338}\right), \quad (12)$$

where the correlation coefficient is 0.9623 between Eq. (12) and data set $[t, R]$, see the curve with the symbol '+' in Fig. 3(b). The curve shows that the fitting with Weibull distribution is not well in this case. To improve the modeling, mixed-Weibull distribution is considered in the form

$$R(t) = p \exp\left[-\left(\frac{t}{\eta_1}\right)^{\beta_1}\right] + (1 - p) \exp\left[-\left(\frac{t}{\eta_2}\right)^{\beta_2}\right], \quad (13)$$

the fitted parameters are $p = 0.2457$, $\eta_1 = 6.3646$, $\beta_1 = 1.1212$, $\eta_2 = 17.2933$, $\beta_2 = 6.0142$ with the correlation coefficient 0.9994 between Eq. (13) and the data set $[t, R]$, see the curve with the symbol within quotes in Fig. 3(b). The mixed-Weibull representation shows little improvement with the proposed (e, c, n) model, Eq. (10), but it takes two additional parameters in the identification.

Many reported data concerned with the failures of cumulative type were studied to check the validity of Eqs. (9a) and (9b) [23], it is found that (e, c, n) model is quite well to show the reliability degradation. For most cases n falls in the range $0.7 < n < 3$. For cumulative damage type failure the memory effect (denoted as the memory characteristics) amplifies the failure possibility with $n < 1$. Under well-controlled experiments in laboratory, e.g. $S-N$ curve [18] or tool wear measurement [22], $n = 1$ is suitable. For this case R can be integrated in a closed form in time domain as

$$R(t) = \frac{e + c}{c + e \exp[(e + c)t]}. \quad (14a)$$

Expand Eq. (14a) in series form as

$$\begin{aligned} R(t) &= \exp[-(e + c)t] \frac{e + c}{c \exp[-(e + c)t] + e} \\ &= \exp\left(-\frac{t}{\theta}\right) \sum_{k=0}^{\infty} f_k(c\theta, e\theta) \left(\frac{t}{\theta}\right)^k, \end{aligned} \quad (14b)$$

where $\theta = 1/(e + c)$. The coefficients f_k are functions of c and e . Eq. (14b) is similar to Erlangian distribution [25] with different expressions of f_k .

3.2. Case 2

The environmental stress screening data [26] were obtained with a set of 141 modules of airborne computer. The reliability and hazard rate are calculated from these

data. Since the data concern with the environmental stress screening, it should be recorded in a burning-in process, i.e. the data are defined in the infant mortality period, thus, the hazard rate depends on the adaptation and white noise. It is assumed that

$$h(R) = e + aR^m. \quad (15)$$

After fitting Eq. (15) becomes

$$h(R) = 0.0002036 + 0.0345106R^{15.42786} \quad (16)$$

with correlation coefficient 0.9184. The comparison of fitted curves with data shown in Fig. 4(b) and (c) indicates that the proposed model is also applicable. Here we would like to mention that the data in this case are only part of failure results, thus the fitted $h(R)$, Eq. (16), cannot be used to describe the overall behavior of the serviced systems which have already past the screening process. However, this relation points out the characteristics of adaptation.

3.3. Case 3

A certain experimental data dealt with a set of 10,000 identical components [27] is shown in Fig. 5. The variation of hazard rate follows a typical bathtub curve. Thus, we assume that

$$h = e + aR^m + c(1 - R)^n. \quad (17)$$

After fitting Eq. (17) becomes

$$h(R) = 0.0007707 + 0.0027899R^{10.9989} + 0.0200885(1 - R)^{6.007695} \quad (18)$$

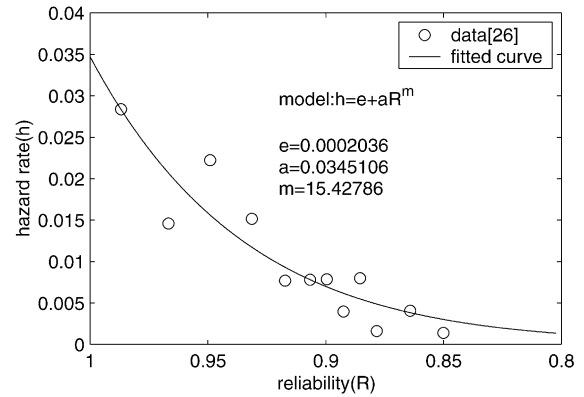
with correlation coefficient 0.988. In this case when the term about man-machine interference effect is added in Eq. (18) the fitted learning coefficient, l , is 1.258×10^{-11} . This means that the improvement due to learning is negligible in this case.

3.4. Case 4

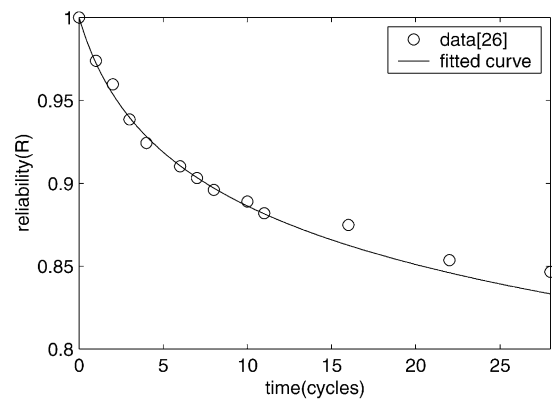
Sometimes a further check of the hazard rate variation, there might be a slight increase (or decrease) at useful life period. One of the reasons is that it comes from the effect of man-machine interference. To clarify this point of view a data set of mortality table is given. At first, h is proposed, without considering the effect of man-machine interference as Eq. (17)

$$h = e + aR^m + c(1 - R)^n.$$

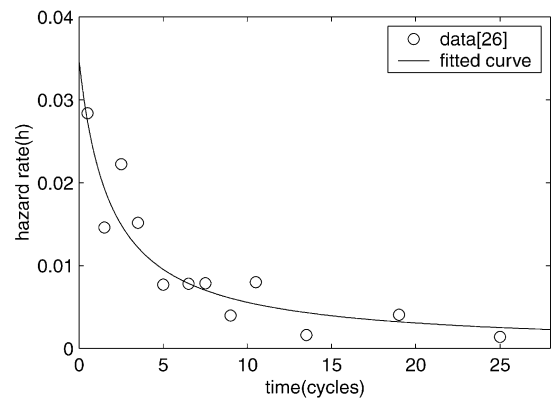
After fitting the estimated parameters are $e = 0.0157598$, $a = 0.7761109$, $m = 12.7859$, $c = 0.219035$, $n = 10.1462$ with correlation coefficient 0.9100 between the fitted function $h(R)$ and data set $[h, R]$. The comparisons of reliability and hazard rate with data are shown in Fig. 6(a) and (b), respectively. The results are not quite well acceptable especially at the middle range of useful life period. To be a more satisfactory fitting, $h(R)$ is further



(a) The fitting of hazard rate vs. reliability



(b) The fitted reliability vs. cycles



(c) The fitted hazard rate vs. cycles

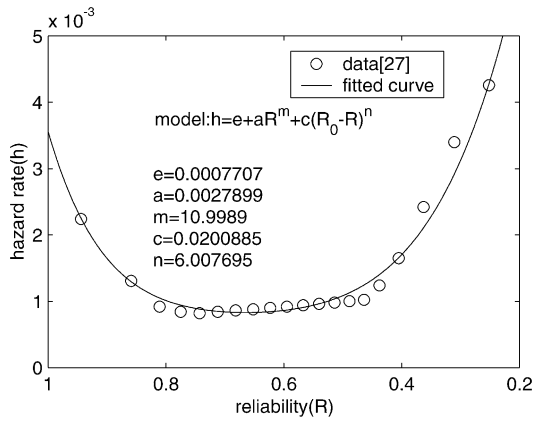
Fig. 4. The comparison of fitted results with data [26] in case 2.

modified as

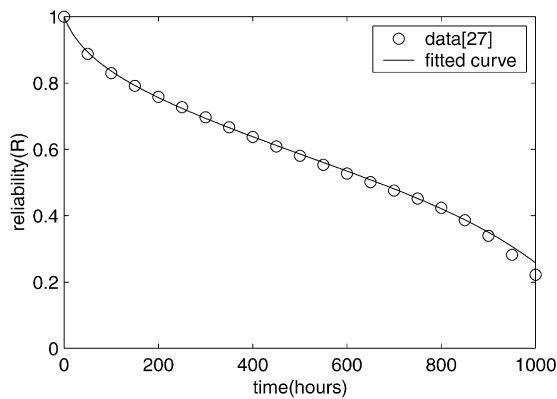
$$h = e + aR^m + c(1 - R)^n + l(1 - R)^{n'}.$$

After fitting Eq. (6b) becomes

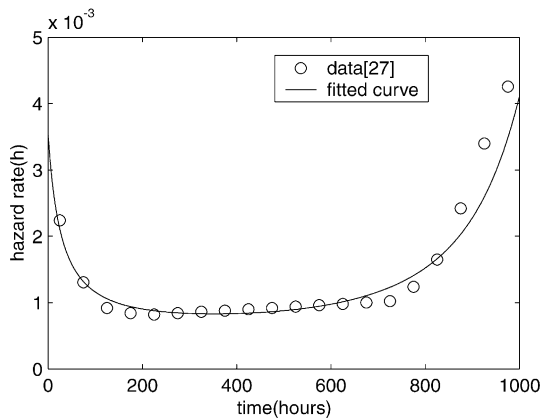
$$h(R) = 6.03090 \times 10^{-6} + 0.60696R^{10.99442} + 0.20347(1 - R)^{13.99175} + 0.07294(1 - R)^{2.22766} \quad (19)$$



(a) The fitting of hazard rate vs. reliability



(b) The fitted reliability vs. time



(c) The fitted hazard rate vs. time

Fig. 5. The comparison of fitted results with data [27] in case 3.

with correlation coefficient 0.9398. By taking one more term the fitting is improved obviously, see Fig. 7. The last term in Eq. (6b), representing the habit-induced failure (a kind of failure with memory characteristics other than (c, n) representation), becomes significant when the system reliability is below 0.6. In a word, this term plays an important role and should be accounted during the system working. Instead of Eq. (6b), Eq. (6a) is also considered in modeling, the result

provides the same accuracy [28] as shown by Eq. (6b). However it is not presented here in the reason that the negative of man-machine interference depends less on the reliability itself due to no correction occurred. Actually from the figure the increase of hazard rate can be approximated by two piecewise straight lines after $t > 20$ years, this points out that at least two different kinds of cumulative damage failure need to be considered.

Since too many parameters are involved in the fitting, a rough estimation of some key parameters can be done first. In our experience the data $1 < R < 0.85$ are used to obtain the initial values of a and m (neglecting the other terms), the data $R < 0.45$ to get the initial values of c and n by keeping the cumulative failure term only. Based on these values, the fittings are carried out trial and error within the whole range of reliability to decide parameters thereafter.

3.5. Case 5

In order to demonstrate the improvement of failure rate due to learning. The hazard rate function is taken as

$$h = e + aR^m + c(1 - R)^n + lR^{m'}(1 - R)^{n'},$$

After fitting Eq. (6a) becomes

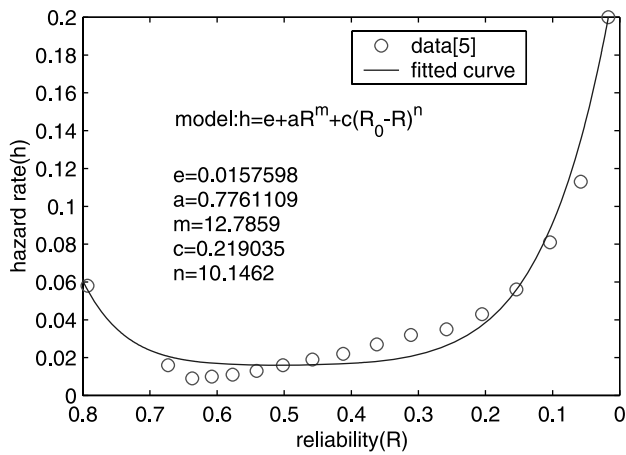
$$\begin{aligned} h(R) = & 0.0020946 + 0.0336211R^{12.63064} \\ & + 0.0263654(1 - R)^{12.3343} \\ & - 0.6951977R^{2.571744}(1 - R)^{9.211067} \end{aligned} \quad (20)$$

with correlation coefficient 0.9978. The comparisons are made in Fig. 8. It shows that the model agrees well with the data. As in case 4 Eq. (6b) is also tested with the same accuracy as Eq. (20) in the modeling [28]. But positive of man-machine interference removes the mistakes and such capability depends on the reliability therefore Eq. (6a) is preferred.

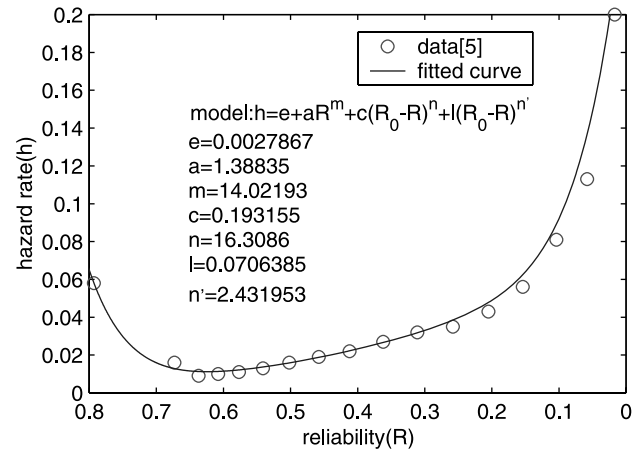
The variations of each item in the hazard rate function of case 4 and case 5 are presented in Fig. 9. The effects of man-machine interference are different. In case 4, this effect is negative (inappropriate operation makes this situation worse) but case 5 positive (learning makes this situation better).

4. Conclusions

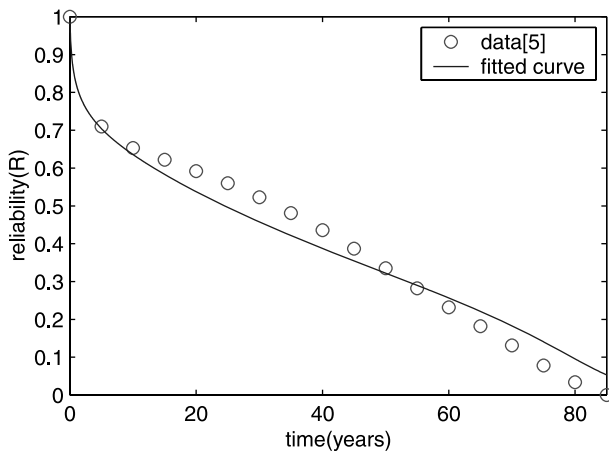
In this paper, the hazard rate function is suggested to be dependent on the reliability. A sophisticated system degrades in many reasons. By assuming each effects the failure behavior independently, the hazard rate function is a summation of the terms based on these mechanisms, including (1) unpredictable failure occurrence due to the intrinsic weakness or/and sudden changes in the environmental conditions, (2) cumulative damage, (3) man-machine interference, and (4) adaptation. The first effect



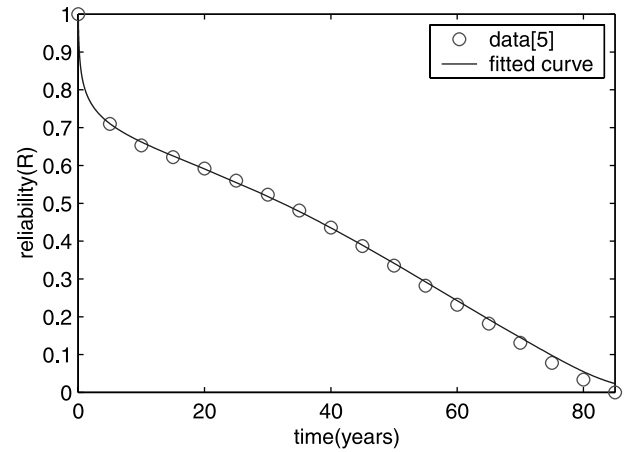
(a) The fitting of hazard rate vs. reliability



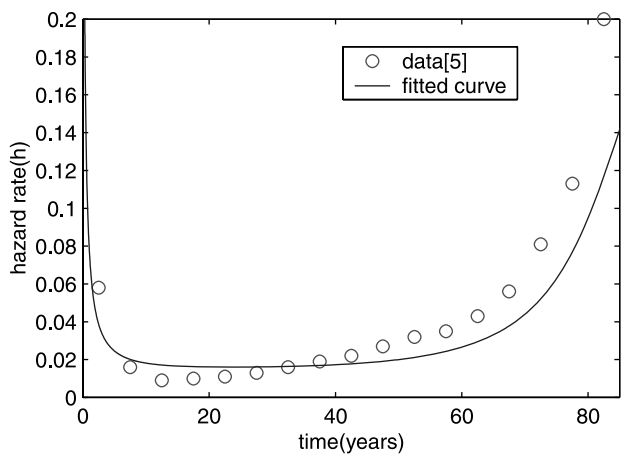
(a) The fitting of hazard rate vs. reliability



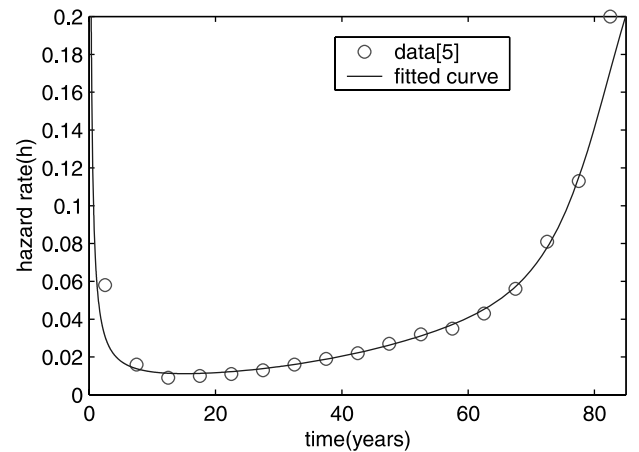
(b) The fitted reliability vs. time



(b) The fitted reliability vs. time



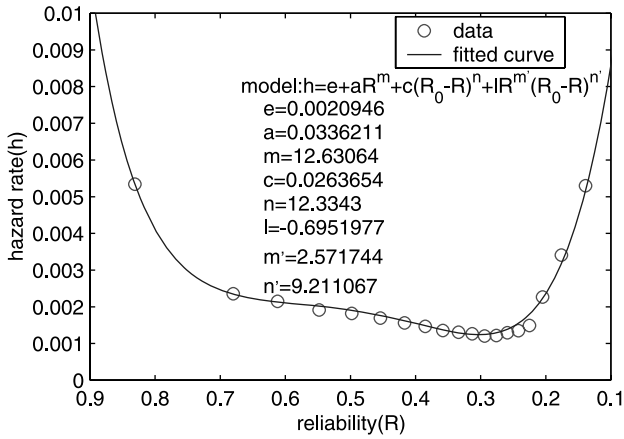
(c) The fitted hazard rate vs. time



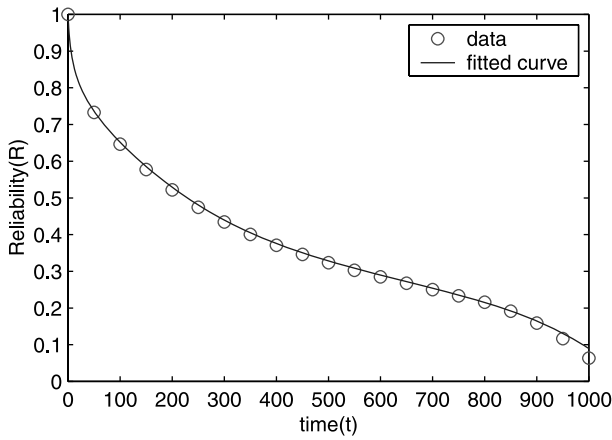
(c) The fitted hazard rate vs. time

Fig. 6. The comparison of fitted results (without the consideration of man-machine interference) with data [5] in case 4.

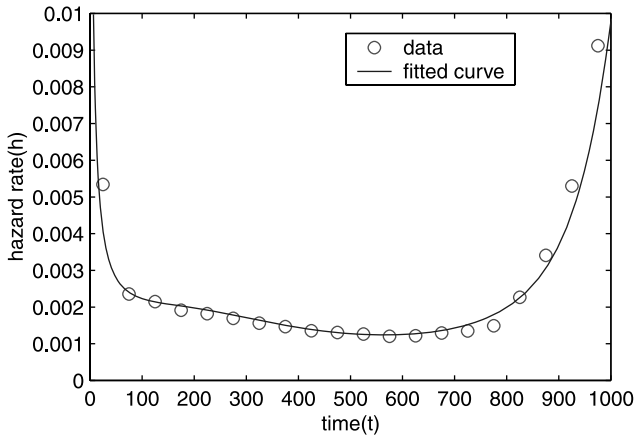
Fig. 7. The comparison of fitted results with data [5] in case 4.



(a) The fitting of hazard rate vs. reliability



(b) The fitted reliability vs. time

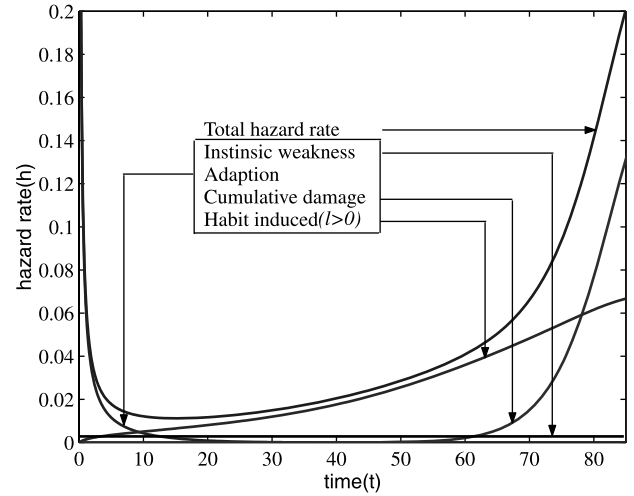


(c) The fitted hazard rate vs. time

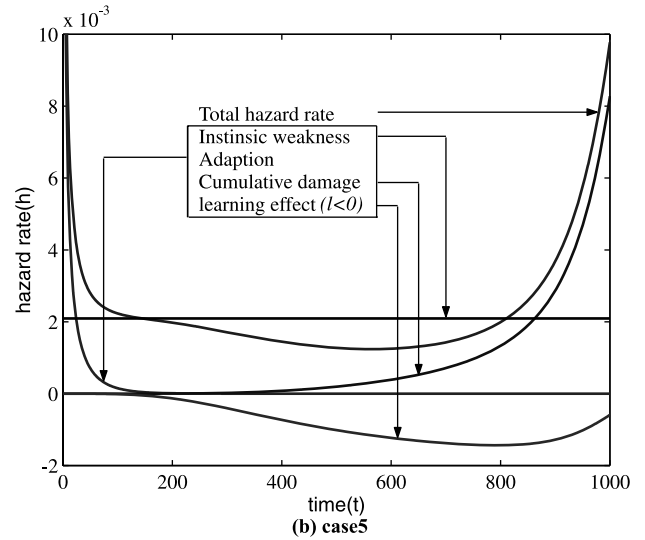
Fig. 8. The comparison of fitted results with raw data in case 5.

can be represented by a constant; the others depend on the reliability, the failure probability or the combination of both. The explicit form is

$$h(R) = e + aR^m + c(1 - R)^n + lR^{m'}(1 - R)^{n'},$$



(a) case4



(b) case5

Fig. 9. Separated item in the hazard rate function.

or

$$h(R) = e + aR^m + c(1 - R)^n + l(1 - R)^{n'}.$$

It is worth mentioning that

1. the proposed model is fitted quite well for existing examples,
2. the fitting procedures are not sophisticated with experience in obtaining the initial guess of key parameters by piecewise fitting of the data $[h, R]$ as discussed in example 4,
3. moreover the discussions of the parameters in this model make sense based on the failure characteristics, and this may lead to some clues in the improvement of design for next generation system.

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